

Global Caffeine Consumption & Health Patterns

Final Presentation & Demo





Introduction

For many, caffeine is part of daily life, and its relationship to stress depends on more than intake alone. We set out to examine how caffeine relates to stress and how factors such as country, age, sleep, and physical activity shape that relationship.

Iteration 1 (Dashboard)

Coffee Health Analytics Dashboard

Interactive analysis of coffee consumption and health metrics



Iteration 2 (Interactive)

Interactive Silhouette (D3) — Coffee Effects on Body

Country
Belgium

Age 35 Gender Female

Caffeine (mg)

Sleep (hours)

Physical Activity (hrs/day) 1 Smoking Yes

Alcohol Consumption Yes

Predicted Stress: 94%

Head color reflects predicted stress (blue → amber → red).
Cup liquid is coffee; particles flow from mouth down the neck and scale with caffeine.



Usability Study

MoSCoW Priority	Issue
Must	Add numeric labels to the sleep slider (e.g., “6 hrs”, “7 hrs”).
Must	Make predicted stress value significantly more prominent (larger font, bold, reposition). Move to the top of the page instead the bottom
Must	Add a color legend/scale explaining head color → stress mapping.
Must	Provide context/explanation of what the stress % represents and how it’s calculated.
Should	Improve title clarity and hierarchy (e.g., bigger title, short subtitle describing purpose).
Should	Add descriptions or tooltips explaining why country and age impact stress.
Should	Enhance visual changes when adjusting certain parameters (e.g., age). Nothing happens when age is changed by itself
Won't	Add message that signals user that new updates have been applied.
Could	Improve right/left layout balance (larger left visualization panel).
Won't	Allow coffee size/type to influence stress more realistically.
Could	Add optional source or “about the data” section.
Could	Add reset button so the viz can return to default values after use.
Won't (for now)	Full redesign of character animation or visual aesthetic.
Must	Correct Title. Says Coffee instead of Caffeine
Must	The color scheme for stress is supposed to be blue -> amber -> red. viz uses green instead of blue. Update note or fix color
Should	Extend color representation for stress onto arms, hand to make it more intuitive.
Must	Stop updating caffeine when selecting new country. Make all countries default to same caffeine (mg).

Interactive Silhouette (D3) — Coffee Effects on Body

Country

USA

Age

35

Gender

Male

Coffee type

Custom

Caffeine (mg)

0 mg

Size (oz)

20 oz

Sleep (hours)

Physical Activity (hrs/day)

1

Smoking

No

Alcohol Consumption

No

Predicted Stress: 15%

Head color reflects predicted stress (blue → amber → red).
Cup liquid is coffee; particles flow from mouth down the neck and scale with caffeine.

Caffeine

Auto update

Hours

Smoker

Stress

Context



Final result

Interactive Silhouette — Caffeine, Stress & Lifestyle Factors

Explore how caffeine intake, sleep, activity, and lifestyle factors influence predicted stress levels.

About the data

The visualization uses a synthetic dataset of about 10,000 profiles inspired by a publicly available coffee and health dataset on Kaggle. The dataset combines self-reported coffee intake with basic health and lifestyle variables such as caffeine intake, cup size, sleep duration, physical activity, smoking and alcohol use, age, gender, country, and self-reported stress. For the interactive model, we use simple, hand-tuned rules anchored on country-level averages from this dataset to estimate stress from your chosen inputs. This is a teaching-focused, exploratory tool only and should not be interpreted as medical advice.

PREDICTED STRESS

15%

Country ⓘ

USA

Age ⓘ

35

Gender ⓘ

Male

Caffeine (mg) ⓘ

0 mg

Sleep (hours) ⓘ

10 hrs

Physical Activity (hrs/day) ⓘ

1

Smoker ⓘ

No

Alcohol Consumption ⓘ

No

What's driving your stress right now?

For a 35-year-old male in USA, the model starts around 34% stress and adjusts based on caffeine, sleep, activity, smoking, alcohol, age, and gender.

Sleep: lowers stress by -17 pts

Caffeine: lowers stress by -5 pts

Age band: raises stress by +2 pts

Head color → Stress level

Low (0%) Mid (50%) High (100%)



Tools Used



D3.js

Core visualization library for interactive charts, animations, and data bindings



HTML/CSS

Responsive layout with CSS Grid and custom styling for the body silhouette



JavaScript

Event handling, DOM manipulation, and real-time updates for user interactions



GitHub Pages

Free hosting platform for deploying our interactive visualization website

Live Demo



Links: [Final Demo](#) | [Final Visualization Site](#)

If we had more time...



Would Add

- Integrate ML model for real stress predictions
- Support multiple coffee types with varying caffeine
- Add enhanced character animations
- Extend stress color to arms and hands



Would Keep

- Human-centered body silhouette design
- Real-time interactive slider controls
- Country comparison functionality
- Smoker vs non-smoker comparison

Contributions

- Abhay Naik: Ideation, Hypothesis Visuals, Iteration 2 Design & Development, Usability, Iteration 3 Design/Updates and Demo Video
- Nichol Flowers: Data Preprocessing, Hypothesis 1 & 2 Analysis + Visualizations, Presentation Design, Usability, Iteration 3 Updates, Site Design & Deployment
- Oleksii Lavrenin: Data Preprocessing, Hypothesis Visualizations, Presentation Design, Usability and Iteration 3 Updates
- Bryan Park: Hypothesis 3 Analysis + Visualizations, Iteration 1 Design & Development, Usability and Iteration 3 updates

The background is a rich, dark brown color. It is decorated with several coffee beans of varying shades (from light tan to dark chocolate) and dynamic splashes of coffee liquid. The splashes are located in the top-left, bottom-left, and bottom-right corners, creating a sense of movement and freshness. The coffee beans are scattered across the frame, some in sharp focus and others blurred, adding depth to the composition.

Thank You

Links: [Final Demo](#) | [Final Visualization Site](#)